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United States Patent Application Publication

20250277407

Kind Code

A1

Publication Date

September 04, 2025

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DEEP LADDER TRAY

Abstract

A tray for a stepladder is disclosed. The tray provides an area for work space or for storage of tools or materials in addition to that provided by the stepladder by itself. The tray is configured to fit within the boundaries of the stepladder when the stepladder is in its stored orientation.

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Appl. No.: 19/211636

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 17130414 20201222 parent-grant-document US 12331592 child US 19211636

Publication Classification

Int. Cl.: E06C1/39 (20060101); E06C1/18 (20060101); E06C7/14 (20060101)

U.S. Cl.:

CPC E06C1/39 (20130101); E06C1/18 (20130101); E06C7/14 (20130101);

Background/Summary

RELATED APPLICATION [0001] This application is a continuation of application Ser. No. 17/130,414, filed Dec. 22, 2020, which is hereby fully incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to a folding tray for use with a ladder that provides for more depth and, thus more volume, but may be conveniently stored within the boundaries of the stepladder when not in use.

BACKGROUND

[0003] Ladders, step stools, and stepladders are generally known in the art. Kummerlin, U.S. Pat. No. 4,502,564, discloses a foldable step ladder. Because of their desired portability, foldable stepladders often include handles and other features that make them easier to transport. Lucci, U.S. Pat. No. 3,744,591, discloses a portable, folding stepladder.

[0004] When working on a ladder, step stool, or stepladder, it is often desirable to have tools, paint, and other necessary objects within easy reach. For example, it is known to removably attach a paint roller tray to the rung of a ladder to more easily paint a ceiling or other area requiring a ladder. Golden, U.S. Pat. No. 3,625,388, discloses a paint tray particularly useful with an upright ladder.

[0005] Utility trays for use with ladder, step stool, or stepladder are also known in the art. Pham, U.S. Pat. No. 5,673,885, discloses a paint tray for a stepladder for storing work materials, tools and a paint bucket that is held onto the stepladder by retaining means. Melanson, U.S. Pat. No. 5,613,574, discloses a stepladder mounted tool holster and parts tray that removably clamps onto the top step of a stepladder. Katz et al., U.S. Pat. No. 6,443,260, discloses a stepladder tray pivotally attached to the top cap of a stepladder for supporting tools and the like. Christ et al., U.S. Pat. No. 5,052,581, discloses a detachable ladder support tray for supporting tools and paint containers. Improvements to add to the amount of space available at the top of a stepladder include U.S. Pat. No. 10,138,680 to Williams, et al., for a hinged tray for a ladder or step stool.

[0006] In addition to utility trays that are removable, there are also trays that are permanently attached to the structure of ladder, step stool, or stepladder. U.S. Pat. No. Williams, et al., 10,138,680, discloses a hinged tray for a ladder or step stool in which the tray is located at the top of the ladder or step stool, and folds to a depth that is not deeper than the folded ladder depth.

[0007] It is often inconvenient, however, to use a removable utility tray with a ladder, step stool or step ladder. In some instances, the tray may be difficult to attach or remove from the ladder, step stool, or stepladder depending on the configuration of the top cap of the ladder, step stool, or stepladder. The removable tray and ladder, step stool, or stepladder usually must be stored separately, taking up additional space. Additionally, to move a ladder, step stool, or stepladder from place to place, the tray may need to be removed and carried separately because of weight or awkward transport configuration. Further, trays located at the top of a ladder, step stool, or stepladder may interfere with use of the ladder, step stool, or stepladder and is often limited in size due to their location.

[0008] There is a need in the industry to have tray incorporated into the structure of a ladder, step stool, or stepladder to hold tools and hardware that can be stored within the boundaries of the ladder, step stool, or stepladder.

SUMMARY

[0009] In embodiments, a folding tray for ladder, step stool, or stepladder is disclosed. Throughout the specification is to be understood that the use of ladder, step stool, or stepladder are interchangeable and the disclosed device may be used with any of the climbing apparatuses. The tray allows for a larger work area on stepladder while the tray can be folded or pivoted to be stored within the boundaries of the closed stepladder. The tray provides a larger and deeper work and storage area than prior art ladder trays without increasing storage space requirements.

[0010] It is an object of the invention to provide an innovative deeper utility tray for use with stepladders. It is also an object of the invention to provide a hinged tray that is integral to the

structure of the stepladder. It is another object of the invention to provide a deep hinged tray that fits within the outer boundaries of the closed stepladder to eliminate any impact on storing of the stepladder from the tray.

[0011] The above summary is not intended to describe each illustrated embodiment or every implementation of the subject matter hereof. The figures and the detailed description that follow more particularly exemplify various embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Subject matter hereof may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying figures, in which:

[0013] FIG. 1 is a rear isometric view of a stepladder in an in-use orientation incorporating an embodiment of the invention.

[0014] FIG. 2 is a front elevation view of the stepladder of FIG. 1.

[0015] FIG. 3 is rear elevation view of the stepladder of FIG. 1.

[0016] FIG. 4 is right side elevation view of the stepladder of FIG. 1.

[0017] FIG. 5 is a left side elevation view of the stepladder of FIG. 1.

[0018] FIG. 6 is a top view of the stepladder of FIG. 1.

[0019] FIG. 7 is a bottom view of the stepladder of FIG. 1.

[0020] FIG. 8 is a rear isometric view of a stepladder in a stored orientation incorporating an embodiment of the invention.

[0021] FIG. 9 is a front elevation view of the stepladder of FIG. 8.

[0022] FIG. 10 is rear elevation view of the stepladder of FIG. 8.

[0023] FIG. 11 is right side elevation view of the stepladder of FIG. 8.

[0024] FIG. 12 is a left side elevation view of the stepladder of FIG. 8.

[0025] FIG. 13 is a top view of the stepladder of FIG. 8.

[0026] FIG. 14 is a bottom view of the stepladder of FIG. 8.

[0027] FIG. 15 is a rear isometric view of an embodiment of the deep ladder tray of the invention.

[0028] FIG. 16 is a front elevation view of the deep ladder tray of FIG. 15.

[0029] FIG. 17 is rear elevation view of the deep ladder tray of FIG. 15.

[0030] FIG. 18 is right side elevation view of the deep ladder tray of FIG. 15.

[0031] FIG. 19 is a left side elevation view of the deep ladder tray of FIG. 15.

[0032] FIG. 20 is a top view of the deep ladder tray of FIG. 15.

[0033] FIG. 21 is a bottom view of the deep ladder tray of FIG. 15.

[0034] While various embodiments are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the claimed inventions to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the subject matter as defined by the claims.

DETAILED DESCRIPTION OF THE DRAWINGS

[0035] Attached are drawings of an embodiment of the deep ladder tray of the present invention incorporated into a stepladder in an in-use orientation (FIGS. 1-7) and in a stored orientation (FIGS. 15-21) as well as isolated drawings of the deep ladder tray itself (FIGS. 8-14). It is understood that the various components disclosed in the drawings may be substituted with equivalent components and are not considered limiting.

[0036] The following detailed description should be read with reference to the drawings in which

similar elements in different drawings are numbered the same. The accompanying figures depict embodiments of the electro-mechanical brake system of the present invention, and features and components thereof. Any references to front and back, right and left, top and bottom, upper and lower, and horizontal and vertical are intended for convenience of description, not to limit the present invention or its components to any one positional or spatial orientation. The drawings, which are not necessarily to scale, depict illustrative embodiments and are also not intended to limit the scope of the invention. Any reference in the claims to a “stepladder” is not intended to limit the scope of the invention to a specific type of climbing device and includes ladders, step stools, etc.

[0037] A stepladder according to an embodiment of the invention is depicted in FIGS. **1-7** in the open or in-use orientation and FIGS. **8-14** in the closed or stored orientation by reference numeral **100**. Stepladder **100** has front rails **110** and rear rails **120** with a hinge **180** between to allow the stepladder **100** to be placed in an open or in-use orientation (FIGS. **1-7**) or in a closed or stored orientation (FIGS. **8-14**). Stepladder **100** typically has steps **130** or rungs or a combination of the two between front rails **110** and potentially between rear rails **120**. Any reference in the claims to steps is not intended to limit the scope of the invention to a specific type of step, but to any type of step, rung, platform, etc. In lieu of steps **130** or rungs between rear rails **120**, cross braces **150** may be used to separate, connect, and support rear rails **120**. Angle braces **160** may be used to further support steps **130** and cross braces **150**. A top cap **145** may be added to the upper portion of the stepladder **100** to provide a work or tool storage surface and to provide a barrier to increase user safety when on the top step **130** or platform **140** of stepladder **100**. Alternatively, instead of a top cap **145**, stepladder **100** may merely have a frame or handles at the top of the rails **110**, **120**.

[0038] Stepladder **100** may also have one or more platforms **140** to provide a more secure standing area when stepladder **100** is in use. Platforms **140** for stepladders **100** are typically folding and rely upon links to move the platform **140** from a stored configuration to an in-use configuration. Stepladders **100** also commonly have a foot **170** on the end of the front rails **110** and rear rails **120** to protect surfaces onto which the stepladder **100** is placed and to provide a wider footprint and better traction for the stepladder **100**.

[0039] In one embodiment, a folding tray **200** is attached to the stepladder **100**. In the embodiment depicted in FIGS. **1-21**, the tray **200** is attached to the stepladder **100** proximate the hinges **180**. In FIGS. **1-7**, the tray **200** is in its use orientation and in FIGS. **8-14** the tray is in its stored orientation. FIGS. **15-21** present illustrations of an embodiment of a tray **200** according to the invention. The tray **200** will typically comprise a front wall **220**, a rear wall **230**, side walls **240**, and a bottom **210**. The tray also has a top surface **205** that is the uppermost extent of the tray **200**. Tray **200** also includes tray shoulders **245** on the tray side walls **240**. Tray **200** may also have various features including a tool apertures **260** for holding tools or materials or a hanging bar **250** for hanging tools or materials. Tool apertures **260** and hanging bar **250** may be within or connected to the tray shoulders **245** as shown in FIGS. **1-7** to the front wall **220**, rear wall **230**, or side walls **240** (not shown).

[0040] Tray **200** has a height dimension **290** that represents the outermost dimension of the tray **200** between the exterior of the front wall **220** and exterior of the rear wall **230** in a front-to-back direction, a width dimension **280** that represents the outermost dimension of the tray **200** between exterior of the shoulders **245**, and a depth dimension **270** that represents the outermost dimension of the tray **200** in a direction between the top surface **205** and exterior of the bottom surface **215**. The combination of these dimensions **270**, **280**, **290** define the outer boundary of the tray **200**. Tray **200** also has a tray width boundary **285**, which represents the outermost dimension of the tray **200** between the exterior of the side walls **240**. The stepladder **100** also includes a platform gap dimension **115**, which represents the distance between the upper edge of the platform **140** and the bottom edge of the top cap **145** when the stepladder **100** is in the stored orientation (FIGS. **8-14**).

[0041] In the embodiment depicted in FIGS. **1-21**, tray **200** forms a single interior storage area between the walls **220**, **230**, **240** and the bottom **210**. However, other embodiments may include

multiple storage areas or work areas by including additional walls (not shown).

[0042] To facilitate storage of the tray **200** when not in use and to increase the amount of storage space provided an embodiment of the tray **200** fits within the boundaries of the stepladder **100** when the stepladder **100** is in the closed or stored orientation. In this embodiment, the tray height dimension **270** is less than the step gap dimension **115**, the tray width dimension **280** is less than the stepladder rail interior width dimension **135** at the location where the tray **200** is in its stored orientation (FIGS. **8-14**), and the tray depth dimension **290** is less than the combined rail depth dimension **155**. A preferred embodiment includes the front rails **110** and rear rails **120** forming a combined rail depth boundary **165** into which the entirety of the tray **200** rests when the stepladder **100** is in the stored orientation. In a further embodiment, the tray shoulder depth dimension **247** is less than the rear rail depth dimension **157** and the tray shoulder width dimension **246** is less than the front rail width dimension **112**. In this configuration the depth of the tray **270** is less than the combined front rail depth dimension **156** and rear rail depth dimension **157**. This arrangement maximizes the overall depth of the tray **200** and allowing tool apertures **260** while the tray **200** can fit within the combined rail depth boundary **165** and ladder width boundary **137** when the stepladder **100** is in the stored orientation. The combined rail boundary **199** represents outer edges of the front rails **110** and rear rails **120** within the platform gap dimension **115** when the ladder **100** is in its stored orientation.

[0043] Various embodiments of systems, devices, and methods have been described herein. These embodiments are given only by way of example and are not intended to limit the scope of the claimed inventions. It should be appreciated, moreover, that the various features of the embodiments that have been described may be combined in various ways to produce numerous additional embodiments. Moreover, while various materials, dimensions, shapes, configurations and locations, etc. have been described for use with disclosed embodiments, others besides those disclosed may be utilized without exceeding the scope of the claimed inventions.

[0044] Persons of ordinary skill in the relevant arts will recognize that the subject matter hereof may comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features of the subject matter hereof may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, the various embodiments can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art. Moreover, elements described with respect to one embodiment can be implemented in other embodiments even when not described in such embodiments unless otherwise noted.

[0045] Although a dependent claim may refer in the claims to a specific combination with one or more other claims, other embodiments can also include a combination of the dependent claim with the subject matter of each other dependent claim or a combination of one or more features with other dependent or independent claims. Such combinations are proposed herein unless it is stated that a specific combination is not intended.

[0046] Any incorporation by reference of documents above is limited such that no subject matter is incorporated that is contrary to the explicit disclosure herein. Any incorporation by reference of documents above is further limited such that no claims included in the documents are incorporated by reference herein. Any incorporation by reference of documents above is yet further limited such that any definitions provided in the documents are not incorporated by reference herein unless expressly included herein.

[0047] For purposes of interpreting the claims, it is expressly intended that the provisions of 35 U.S.C. § 112(f) are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1. A ladder tray system comprising: a ladder having an in-use orientation and a stored orientation comprising: a front rail comprising a front rail width dimension and a front rail depth dimension; a rear rail substantially aligned with the front rail in the in-use orientation comprising a rear rail depth dimension; a step; a platform; and a tray comprising: a tray width dimension; a tray height dimension; a tray depth dimension; and a tray shoulder comprising a tray shoulder depth dimension and a tray shoulder width dimension, the tray shoulder depth dimension being less than the tray depth dimension; a ladder width boundary; a combined rail depth boundary; and wherein the tray width dimension fits within the ladder width boundary when the ladder is in the stored orientation; wherein the tray depth dimension substantially fits within the combined rail depth boundary when the ladder is in the stored orientation; and wherein the tray shoulder depth dimension is less than or equal to the rear rail depth dimension.
2. The ladder tray system of claim 1 wherein the tray shoulder width dimension is substantially the same as the front rail width dimension.
3. The ladder tray system of claim 2 further comprising a platform gap dimension and wherein the tray height dimension fits within the platform gap dimension when the ladder is in the stored orientation.
4. The ladder tray system of claim 2 wherein the shoulder comprises a tool aperture.
5. The ladder tray system of claim 2 further comprising a hanging bar.
6. The ladder tray system of claim 2 further comprising a link between the tray and platform wherein pivoting the platform with respect to the front rail or rear rail results in pivoting of the tray.
7. The ladder tray system of claim 6 further comprising a tray bottom surface wherein the tray bottom surface is substantially parallel to the rear rail when the ladder is in the stored orientation.
8. The ladder tray system of claim 6 wherein the tray rests entirely within a combined rail depth boundary when the ladder is in the stored orientation.
9. The ladder tray system of claim 2 wherein the tray depth dimension is larger the rear rail depth dimension and larger than the front rail depth dimension.
10. The ladder tray system of claim 2 further comprising a rail interior width dimension wherein the wherein the tray width boundary fits within the rail interior width dimension when the ladder is in the stored orientation.
11. The ladder tray system of claim 3 wherein the tray depth dimension is larger the rear rail depth dimension and larger than the front rail depth dimension.
12. A tray for use in combination with a ladder, the tray comprising: a tray width dimension; a tray height dimension; a tray depth dimension; and a tray shoulder comprising a tray shoulder depth dimension and a tray shoulder width dimension, the tray shoulder depth dimension being less than the tray depth dimension; wherein the tray has an in-use orientation and a stored orientation and is pivotably attached to a ladder, the ladder comprising: a front rail comprising a front rail width dimension and a front rail depth dimension; a rear rail generally aligned with the front rail in the in-use orientation comprising a rear rail depth dimension; a step; and a platform; wherein the tray shoulder depth dimension is less than or equal to the rear rail depth dimension and wherein the tray rests substantially within a combined rail depth boundary when the tray is in the stored orientation.
13. The of claim 12 further comprising a platform gap dimension and wherein the tray height dimension fits within the platform gap dimension when the tray is in the stored orientation.
14. The tray of claim 12 wherein the ladder further comprises: a ladder width boundary; a combined rail depth boundary; and wherein the tray width dimension fits within the ladder width boundary when the tray is in the stored orientation; and wherein the tray depth dimension fits within the combined rail depth boundary when the tray is in the stored orientation.
15. The tray of claim 14 wherein the tray rests entirely within a combined rail depth boundary

when the tray is in the stored orientation.

16. The tray of claim 15 wherein the tray depth dimension is larger than the rear rail depth dimension.

17. The tray of claim 16 wherein the tray depth dimension is larger than the front rail depth dimension.

18. The ladder tray system of claim 12 wherein the shoulder comprises a tool aperture.

19. The ladder tray system of claim 16 further comprising a link between the tray and platform wherein pivoting the platform with respect to the front rail or rear rail results in pivoting of the tray.

20. A ladder having an in-use orientation and a stored orientation comprising: a front rail comprising a front rail width dimension and a front rail depth dimension; a rear rail generally lined up with the front rail in the in-use orientation comprising a rear rail depth dimension; a step; a tray comprising: a tray width dimension; a tray height dimension; a tray depth dimension; and a tray shoulder comprising a tray shoulder depth dimension and a tray shoulder width dimension, the tray shoulder depth dimension being less than the tray depth dimension; a ladder width boundary; a combined rail depth boundary; and wherein the tray width dimension fits within the ladder width boundary when the ladder is in the stored orientation; wherein the tray depth dimension fits within the combined rail depth boundary when the ladder is in the stored orientation; wherein the tray shoulder depth dimension is less than or equal to the rear rail depth dimension; and wherein the tray shoulder width dimension is less than the ladder width boundary.
